BIO 005 Human Physiology · Week 7 notes · Dr. Sharilyn Rennie

The neuromuscular junction and contraction

The written notes for the the neuromuscular junction and contraction walkthrough. Week 7 competencies 1 to 5.

To print: these notes as a two-column PDF. Already printed the Week 7 notes? You do not need to print these again.

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Step by step: from command to contraction

In your textbook: Silverthorn Chapter 12, the part on the somatic motor division and the neuromuscular junction, and Chapter 13, the skeletal muscle sections.

At the neuromuscular junction, step by step

1. A somatic motor neuron, with its cell body in the spinal cord or brainstem, fires an action potential.
2. The action potential travels down the axon and its branches to an axon terminal on one muscle fiber.
3. It opens voltage-gated Ca2+ channels in the membrane of the axon terminal. Ca2+ flows in from the extracellular fluid.
4. The Ca2+ triggers docked synaptic vesicles to fuse with the membrane and release acetylcholine (ACh) into the synaptic cleft by exocytosis.
5. ACh diffuses across the cleft. Two ACh molecules bind to each nicotinic ACh receptor on the motor end plate, and its channel opens.
6. The channel is a nonspecific cation channel. Na+ flows in and a smaller amount of K+ flows out, so the end plate depolarizes: the end plate potential.
7. The end plate potential always reaches threshold (the safety factor). Voltage-gated Na+ channels next to the end plate open, and a muscle action potential spreads along the sarcolemma in both directions and down the T tubules.
8. Acetylcholinesterase in the cleft breaks ACh into acetate and choline. The receptor channels close, and the terminal takes the choline back in to make new ACh.

Excitation-contraction coupling, step by step

1. The muscle action potential travels down a T tubule, deep into the fiber.
2. It passes a DHP receptor, a voltage sensor in the T tubule membrane, which changes shape.
3. In skeletal muscle, the DHP receptor is physically linked to the ryanodine receptor (RyR1), a Ca2+ release channel in the membrane of the terminal cisterna. The shape change pulls it open.
4. Ca2+ flows out of the sarcoplasmic reticulum into the cytosol around the myofibrils. No Ca2+ from outside the cell is needed.

Calcium and the thin filament, step by step

1. Ca2+ binds to troponin C, part of the troponin complex on the thin filament.
2. Troponin changes shape and pulls tropomyosin aside.
3. The myosin binding sites on actin are uncovered.
4. Myosin heads bind firmly to actin and the crossbridge cycle runs for as long as Ca2+ stays bound.

One turn of the crossbridge cycle, step by step

1. Binding: the cocked myosin head, holding ADP and Pi, binds to actin.
2. Power stroke: release of Pi triggers the power stroke. The head swings and pulls the thin filament toward the M line.
3. At the end of the power stroke the head releases ADP and stays tightly bound to actin, with nothing on it. This is the rigor state.
4. Detachment: ATP binds to the head, and the head lets go of actin.
5. Reactivation: the head splits the ATP into ADP and Pi, and the energy re-cocks the head. It holds the ADP and Pi until it binds actin again.

Relaxation, step by step

1. The motor neuron stops firing, and acetylcholinesterase clears the ACh from the cleft.
2. No more muscle action potentials, so the ryanodine receptors close.
3. SERCA pumps in the SR membrane use ATP to move Ca2+ from the cytosol back into the SR.
4. Ca2+ comes off troponin, and tropomyosin slides back over the myosin binding sites.
5. The crossbridges stop. The sarcomeres return to resting length as the muscle is stretched by gravity or an opposing muscle and as titin springs back.

How a skeletal muscle is built

In your textbook: Silverthorn Chapter 13, the part on skeletal muscle anatomy.

- Skeletal muscles are attached to bones by tendons and are controlled by somatic motor neurons.
- A whole muscle is wrapped in connective tissue and divided into fascicles, bundles of muscle fibers.
- A muscle fiber is one muscle cell: very long, with many nuclei, formed when many embryonic cells fused.

- Most of a fiber is filled with myofibrils, rods of contractile proteins that run the length of the cell.
- Each myofibril is a chain of sarcomeres, the contractile units, lined up end to end.

Sarcolemma
The cell membrane of a muscle fiber.

T tubules (transverse tubules)
Narrow tunnels of sarcolemma that dip deep into the fiber. They are open to the extracellular fluid and carry the action potential inside.

Sarcoplasmic reticulum (SR)
The muscle fiber's endoplasmic reticulum. A sleeve of tubes around each myofibril that stores Ca2+.

Terminal cisternae
The swollen ends of the SR, lying right beside a T tubule.

Triad
One T tubule with a terminal cisterna on each side.

Thick filament
Made of myosin. Its heads stick out toward the thin filaments around it.

Thin filament
Made mostly of actin, with tropomyosin and troponin lying along it. Anchored to a Z disc.

Titin
A huge elastic protein that runs from a Z disc to the M line. It holds the thick filaments in place and springs back after a stretch.

Z disc
The boundary at each end of a sarcomere. Thin filaments attach to it.

M line
The center of the sarcomere, where the thick filaments are held together.

Band	What is in it	When the sarcomere shortens
A band (dark)	The full length of the thick filaments, overlapping the thin filaments near its ends	Stays the same length
I band (light)	Thin filaments only, split by a Z disc	Gets shorter
H zone	The middle of the A band: thick filaments, no thin filaments	Gets shorter or disappears

- Sliding filament theory: neither filament gets shorter. Myosin heads pull the thin filaments toward the M line, so the filaments overlap more and the Z discs move closer together.
- The alternating bands are why skeletal muscle is called striated muscle.

At the neuromuscular junction

In your textbook: Silverthorn Chapter 12, the part on the somatic motor division and the neuromuscular junction.

Neuromuscular junction
The synapse between a somatic motor neuron and a skeletal muscle fiber: the axon terminal, the synaptic cleft and the motor end plate.

Motor end plate
The folded patch of sarcolemma under the axon terminal, packed with nicotinic ACh receptors.

Nicotinic ACh receptor
A channel that opens when two ACh molecules bind. It is a nonspecific cation channel: Na+ in exceeds K+ out, so the end plate depolarizes.

End plate potential
The depolarization of the motor end plate. It is a graded potential that happens only on the end plate.

Safety factor
In a healthy junction, the end plate potential is much larger than needed to reach threshold, so every motor neuron action potential produces one muscle action potential.

- Somatic motor neurons always release ACh and always excite the muscle fiber. No neuron inhibits a skeletal muscle fiber directly.
- Na+ has a much larger driving force than K+ at the resting potential, which is why more Na+ comes in than K+ goes out.
- Brain and spinal cord synapses usually need many inputs to add together to reach threshold. The neuromuscular junction is one for one.

What blocks it	Where it acts	How far the signal gets	The muscle
Botulinum toxin	Destroys the proteins vesicles need to fuse, inside the axon terminal	Action potential arrives and Ca2+ enters, but no ACh is released	Limp, flaccid paralysis
Curare	Binds the nicotinic receptors without opening them	ACh is released but cannot bind; no end plate potential	Limp, flaccid paralysis
Myasthenia gravis	Antibodies block and destroy nicotinic receptors	End plate potential is smaller and fails during repeated firing	Weakness that worsens with use and recovers with rest

What this means for patients

- Botulism from contaminated food causes dangerous paralysis. Tiny doses of botulinum toxin (Botox) treat some muscle spasms and chronic migraine and smooth wrinkles.
- Drugs related to curare relax the muscles during surgery while a machine breathes for the patient.
- Myasthenia gravis: during steady firing, a little less ACh is released with each action potential. With fewer receptors, later end plate potentials fall below threshold. Drooping eyelids and double vision are often the first signs.
- Myasthenia gravis is treated with acetylcholinesterase inhibitors, which let ACh stay in the cleft longer, and with treatments that calm the immune attack.

Excitation-contraction coupling

In your textbook: Silverthorn Chapter 13, the part on excitation-contraction coupling.

Excitation–contraction coupling
: The link between the muscle action potential (excitation) and the sliding of the filaments (contraction). The link is Ca2+ released from the SR.

DHP receptor
: A voltage sensor in the T tubule membrane, named for the dihydropyridine drugs that bind to it.

Ryanodine receptor
: A Ca2+ release channel in the SR membrane, named for ryanodine, a plant compound that binds to it.

	Skeletal muscle	Heart muscle
DHP receptor	A voltage sensor physically linked to the ryanodine receptor	A Ca2+ channel that lets Ca2+ in from outside the cell
What opens the ryanodine receptor	A mechanical pull from the DHP receptor	Ca2+ that came in through the DHP channel: calcium-induced calcium release
Ca2+ from outside the cell needed?	No	Yes
Without extracellular Ca2+	Still contracts	Does not contract

- T tubules carry the action potential deep into the wide fiber, right next to the terminal cisternae, so the whole fiber responds at once.

Calcium and the regulatory proteins

In your textbook: Silverthorn Chapter 13, the part on troponin, tropomyosin and the role of calcium in contraction.

- Each thin filament is two chains of actin twisted together. Every actin molecule has a myosin binding site.
- At rest, tropomyosin lies over the binding sites, and troponin holds it there. Myosin heads can only bind weakly.
- Ca2+ binds troponin C. Troponin changes shape and pulls tropomyosin off the binding sites. Myosin binds firmly and the power strokes begin.
- Ca2+ never binds actin or myosin here. Its target is troponin.

If this fails	Ca2+ in the cytosol	Crossbridges	The muscle feels
Ryanodine receptors will not open (nerve, end plate and ATP normal)	None released	Cannot bind; ATP keeps heads free	Limp
No ATP	Leaks out of the SR	Locked onto actin	Stiff (rigor)

The crossbridge cycle

In your textbook: Silverthorn Chapter 13, the part on the crossbridge cycle and the molecular basis of contraction.

Stage	What happens	What is on the head after it
1. Binding	The cocked head binds to actin	ADP and Pi
2. Power stroke	Pi release triggers the power stroke; the thin filament moves toward the M line; then ADP is released	Nothing (rigor state)
3. Detachment	ATP binds, and the head lets go of actin	ATP
4. Reactivation	ATP is split into ADP and Pi, and the energy re-cocks the head	ADP and Pi

- ATP binds at detachment. ATP is hydrolyzed at reactivation. These are two different stages.
- The thin filament moves only during the power stroke.
- Each turn of the cycle uses one ATP per head. Heads along a thick filament cycle at different times, so the filament does not slip back.
- Without ATP, the heads finish the power stroke and stay locked onto actin: the rigor state. The muscle is stiff, not limp.

What this means for you

- Rigor mortis: after death, ATP runs out, Ca2+ leaks out of the SR, and the myosin heads lock onto actin. The body stays stiff until the muscle proteins begin to break down.
- Relaxation needs ATP to release each head from actin.

Relaxation

In your textbook: Silverthorn Chapter 13, the part on relaxation, and Chapter 12, the part on the neuromuscular junction.

Shutoff job	Where	How	Uses ATP?	If it fails
Acetylcholinesterase (AChE)	Synaptic cleft	Splits ACh into acetate and choline; receptor channels close	No	ACh piles up: twitching, then weakness and paralysis
SERCA (sarco-endoplasmic reticulum Ca2+-ATPase)	SR membrane	Pumps Ca2+ from the cytosol back into the SR, against its gradient	Yes	Ca2+ stays on troponin: the muscle stays contracted and cannot relax

- Clearing ACh stops new action potentials, but the muscle relaxes only when SERCA has removed the Ca2+.
- Relaxation costs energy in two places: ATP to release each myosin head, and ATP for SERCA.

What this means for patients

- Organophosphates, in some insecticides and in nerve agents such as sarin, block acetylcholinesterase.
- At the neuromuscular junction, extra ACh first causes repeated firing and twitching. As the end plate stays depolarized, the nearby voltage-gated Na+ channels become inactivated, and the muscles become weak and paralyzed, including the breathing muscles.
- ACh also builds up at autonomic synapses: small pupils, drooling, sweating and a slow heart rate.

- Treatment includes atropine, which blocks the effects of extra ACh on the organs but does not help muscle weakness, and pralidoxime, which can restore acetylcholinesterase if given early.
- Brody disease, a rare inherited condition in which skeletal muscle SERCA works poorly, makes muscles stiffen and relax slowly after exercise.

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Twitches, summation and motor units

The written notes for the twitches, summation and motor units walkthrough. Week 7 competencies 6 to 10, with part of 16.

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Step by step: from one shock to a full contraction

In your textbook: Silverthorn Chapter 13, the parts on the muscle twitch, summation and tetanus, and motor units.

One twitch, step by step

1. A single stimulus, from the motor neuron or from a stimulator, fires one action potential in the muscle fiber. It lasts only a few milliseconds.
2. Latent period: the action potential spreads down the T tubules, the Ca2+ release channels of the sarcoplasmic reticulum open, Ca2+ binds troponin, tropomyosin moves aside, and crossbridges begin to form. Tension has not risen yet.
3. Contraction phase: Ca2+ is high, more and more crossbridges cycle, and tension rises to a peak.
4. The Ca2+ pump in the sarcoplasmic reticulum membrane, SERCA, uses ATP to pull Ca2+ back out of the cytosol. Ca2+ starts to fall before tension peaks.
5. Relaxation phase: as Ca2+ leaves troponin, tropomyosin covers the binding sites again, crossbridges let go, and tension falls back to zero. This is the longest phase.

Summation and tetanus, step by step

1. A second action potential arrives before the fiber has fully relaxed.
2. Some Ca2+ from the first one is still in the cytosol, because SERCA has not finished pumping it back.
3. The second release adds to what is left over, so Ca2+ rises higher than after one action potential.
4. More troponin has Ca2+ bound, more crossbridges pull at once, and tension climbs above a single twitch: summation.
5. At a higher frequency, the pumps fall further behind. Tension climbs and dips with each stimulus: unfused tetanus.
6. At the highest frequency, Ca2+ stays high between stimuli and tension reaches a smooth plateau: fused tetanus, the strongest contraction the fiber can make.
7. The action potentials themselves never add together. They are all-or-none and have a refractory period.

Recruiting motor units, step by step

1. The brain or a reflex sends excitatory input to the pool of motor neurons for one muscle in the ventral horn of the spinal cord.
2. The smallest motor neurons reach threshold first, because they need the least synaptic input. They fire their small motor units of slow oxidative fibers. The force is small and finely controlled.
3. As the input grows, medium motor neurons reach threshold and add their fast oxidative-glycolytic fibers.
4. With still more input, the largest motor neurons fire and add their many fast glycolytic fibers. The force jumps.
5. Motor neurons that are already active also fire faster, so their twitches summate toward tetanus.
6. In a long steady contraction, the nervous system takes turns among motor units so some rest while others work.

Lifting a load, step by step

1. The muscle is stimulated and tension starts to rise. At first the muscle does not change length.
2. If the load is smaller than the most tension the muscle can make, tension rises until it equals the load.
3. Then the muscle shortens and moves the load, with tension steady at the level of the load: a concentric isotonic contraction.
4. If the load is more than the muscle can make, tension rises to its maximum and the muscle stays the same length: an isometric contraction.
5. If the load is a little more than the tension the muscle is making, the active muscle is stretched while it keeps pulling: an eccentric contraction.

The muscle twitch

In your textbook: Silverthorn Chapter 13, the part on the muscle twitch.

A twitch is a single contraction and relaxation cycle of a muscle fiber, produced by a single action potential. A twitch tracing plots tension against time, with the stimulus marked on the time axis.

Phase	What is happening	What limits how long it lasts
Latent period	Tension is flat. The action potential spreads into the T tubules, Ca2+ is released from the sarcoplasmic reticulum, binds troponin, and tropomyosin moves aside	Excitation-contraction coupling, the time it takes to get Ca2+ onto troponin and crossbridges cycling
Contraction phase	Tension rises to a peak as more crossbridges cycle	How long Ca2+ stays high, and how fast the crossbridges cycle
Relaxation phase	Tension falls back to zero	How fast SERCA pumps Ca2+ back into the sarcoplasmic reticulum. It is the longest phase

- The muscle action potential lasts only a few milliseconds and is finished before tension even starts to rise.
- Tension follows the Ca2+ in the cytosol, not the action potential. Ca2+ starts falling before tension peaks.
- If SERCA is slowed, Ca2+ stays high longer, the relaxation phase gets much longer and the peak a little higher. The latent period does not change.
- Relaxation costs energy: SERCA uses ATP to move Ca2+ against its gradient.
- Because the electrical event is so short and the contraction so long, a second action potential can arrive while the fiber is still contracting. That makes summation possible.

Twitch

One contraction and relaxation of a fiber in response to one action potential.

Latent period

The delay between the muscle action potential and the start of tension, while excitation-contraction coupling happens.

SERCA

The Ca2+ ATPase in the sarcoplasmic reticulum membrane that pumps Ca2+ out of the cytosol and ends a contraction.

Summation and tetanus

In your textbook: Silverthorn Chapter 13, the part on summation and tetanus.

The force of one fiber is graded by how often it is stimulated. The faster the stimuli come, the less time SERCA has to clear Ca2+ between them.

Stimulus frequency	Ca2+ between stimuli	Tension
Low	Falls all the way back to rest	Separate twitches
A second stimulus during relaxation	Some is left over, so the second release adds to it	Summation: the second contraction climbs higher than one twitch
Higher	Never falls to rest, rises and falls with each stimulus	Unfused (incomplete) tetanus: tension climbs and dips with each stimulus
Highest	Stays high	Fused (complete) tetanus: a smooth plateau, the most tension the fiber can make

- Summation comes from Ca2+ building up in the cytosol and more crossbridges pulling at once. It does not come from action potentials adding together; they are all-or-none and have a refractory period.
- In fused tetanus, the tension stops dipping before the Ca2+ does. Ca2+ still rises and falls a little with each stimulus, but never drops low enough for many crossbridges to let go.
- Everyday movements use trains of action potentials and tetanic contractions, not single twitches.
- The disease tetanus is something else: a toxin from Clostridium tetani blocks inhibitory neurons in the spinal cord, so motor neurons fire without control and muscles lock in spasm.

Summation

A contraction added onto one that has not finished relaxing, giving more tension than a single twitch.

Unfused tetanus

Tension that stays raised but rises and falls with each stimulus. Also called incomplete tetanus.

Fused tetanus

A smooth, maximal plateau of tension at a high stimulus frequency. Also called complete tetanus.

Motor units and recruitment

In your textbook: Silverthorn Chapter 13, the part on motor units.

A motor unit is one somatic motor neuron and all the skeletal muscle fibers it innervates. Each fiber is innervated by only one motor neuron, but one motor neuron branches to many fibers.

- When the motor neuron fires, every fiber in its unit gives a full twitch, and no other fibers do.
- The fibers of one unit are spread through the muscle, not bunched together, so even one small unit pulls on the whole muscle evenly.
- All the fibers in one motor unit are the same fiber type.
- Muscles for fine, precise movement, such as the eye muscles and the small muscles of the hand, have small motor units with only a few fibers each. Each added unit adds a small step of force.
- Muscles for gross movement and support, such as the calf muscle, have large motor units with hundreds to thousands of fibers each. Each added unit adds a big step of force.

Motor neuron	Its fibers	Recruited
Small	A few slow oxidative fibers: slow, fatigue-resistant, small force	First, for light efforts
Medium	More fast oxidative-glycolytic fibers: fast, fairly fatigue-resistant	As effort increases
Large	Many fast glycolytic fibers: fast, the most force, tire quickly	Last, for the heaviest efforts

Recruitment

Adding more motor units to increase the force of a whole muscle.

Size principle

Motor units are recruited in order of size, smallest first, because small motor neurons reach threshold with less synaptic input.

Asynchronous recruitment

Taking turns among motor units during a long steady contraction, so some rest while others work and the muscle is slower to fatigue.

The nervous system grades the force of a whole muscle in two ways: how many motor units are active (recruitment) and how fast each one fires (summation and tetanus). Picking up a paper cup uses only small units; lifting a gallon of milk adds medium and then large units.

Stimulus voltage in the lab (Competency 16)

- In PhysioEx Exercise 2 a stimulator shocks a whole muscle, and you raise the voltage step by step.
- Below threshold, no fibers respond and there is no twitch.
- Threshold stimulus: the lowest voltage that produces a visible twitch, because only a few fibers reach threshold.
- Between threshold and maximal, more fibers respond with each step, so the twitch gets stronger. Each fiber still follows the all-or-none rule.
- Maximal stimulus: the lowest voltage at which every fiber responds. Higher voltages add no more force.
- Raising the voltage on a whole muscle works much like recruiting motor units in the body: force grows because more fibers take part.

Length and tension

In your textbook: Silverthorn Chapter 13, the part on the length-tension relationship.

The active tension a fiber makes depends on the length of its sarcomeres before it contracts, because crossbridges form only where myosin heads overlap thin filaments.

Sarcomere length	Filaments	Active tension
Too short (about 1.5 micrometers)	Thin filaments from the two sides overlap each other and get in each other's way; thick filaments press against the Z discs	Lower
Best (about 2.0 to 2.2 micrometers)	Every myosin head can reach a thin filament, and the thin filaments have room to slide	Greatest
Too long (about 3.2 micrometers)	Thin filaments barely overlap the ends of the thick filaments, so few heads can reach actin	Lower, and zero once there is no overlap

- Tension falls at both ends of the curve, for different reasons: too short, the filaments jam; too long, there is too little overlap.
- The 2.0 to 2.2 micrometer plateau in the textbook curve comes from classic measurements of frog muscle. Human thin filaments are a little longer, so human muscle works best at a slightly longer sarcomere length.
- In the body, the way muscles attach to bones keeps their resting length close to the best overlap.
- Bend your wrist all the way forward and make a fist: the finger flexors are already short, and the grip is weaker.
- A muscle pulled far past its resting length can form few crossbridges and makes only a little force.

Length–tension relationship

The active tension a sarcomere makes depends on its length when it is stimulated, and is greatest at the length with the best filament overlap.

Isometric and isotonic contraction

In your textbook: Silverthorn Chapter 13, the part on the mechanics of body movement, isotonic and isometric contractions.

Contraction	Muscle length	Tension compared with the load	Example
Isotonic, concentric	Shortens	Equal to the load, which is less than the muscle's maximum	Curling a dumbbell up
Isometric	Stays the same	Equal to the load (holding still), or the load is more than the muscle can make	Holding a weight still; pushing on a wall
Eccentric	Lengthens while active	A little less than the load	Lowering a dumbbell slowly; walking downstairs

- In every contraction that moves a load, tension first rises until it equals the load, with no change in length, and only then does the muscle shorten.
- An isometric contraction uses ATP the whole time, even though nothing moves.
- In an isometric contraction the sarcomeres shorten slightly as they stretch the tendons and connective tissue, the series elastic elements, but the muscle as a whole does not change length.
- Eccentric work, such as walking downhill or lowering weights, is the kind most likely to cause delayed onset muscle soreness a day or two later.

The force-velocity relationship

- With the muscle fully activated, the speed of shortening falls as the load rises.
- With almost no load, the muscle shortens fastest.
- When the load equals the most tension the muscle can make, the speed is zero and the contraction is isometric.
- This is why you can throw a light ball much faster than a heavy one.

Isotonic contraction

A contraction that moves a load. Concentric when the muscle shortens, eccentric when the active muscle lengthens.

Isometric contraction

A contraction that makes tension without changing the length of the muscle.

Force–velocity relationship

The heavier the load, the slower a muscle shortens; at the muscle's maximum tension, shortening stops.

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Fuel, fiber types and fatigue

The written notes for the fuel, fiber types and fatigue walkthrough. Week 7 competencies 11 to 15.

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Step by step: from fuel to fatigue and recovery

In your textbook: Silverthorn Chapter 13, the parts on muscle energy, fatigue and the classification of muscle fibers.

Where a working fiber spends ATP

1. A motor neuron fires, and an action potential spreads along the fiber's membrane and down the T tubules.
2. Ca2+ is released from the sarcoplasmic reticulum and lets the crossbridges form.
3. Each myosin head splits one ATP on every turn of the crossbridge cycle.
4. The SERCA pump uses ATP to move Ca2+ back into the sarcoplasmic reticulum, so the fiber can relax.
5. The Na+/K+ pump uses ATP to restore the Na+ and K+ gradients after the action potentials.
6. The fiber stores only a few seconds' worth of ATP, so it has to make more as fast as it spends it.

An all-out sprint that becomes a long run, source by source

1. In the first seconds, the ATP already in the fiber is used, and creatine kinase moves a phosphate from creatine phosphate onto ADP to make more ATP in one step. No oxygen is needed.
2. Within roughly 10 to 15 seconds, most of the creatine phosphate is used up.
3. Anaerobic glycolysis carries most of the load next. In the cytosol, glucose is split into two pyruvate, making a net 2 ATP (3 when the glucose comes straight off stored glycogen).
4. Because oxygen cannot be delivered fast enough, pyruvate is turned into lactate, which regenerates NAD+ so glycolysis can continue.
5. After about a minute, glycolysis falls off: H+ builds up and slows its enzymes, and oxidative phosphorylation takes over more of the work.
6. Meanwhile breathing, heart rate and blood flow to the muscle rise, and oxidative phosphorylation in the mitochondria comes up to speed.
7. At a steady pace, oxidative phosphorylation supplies nearly all the ATP, about 30 to 32 ATP in all from each glucose, and can also burn fatty acids. It lasts as long as oxygen and fuel last.

Fatigue during a long steady grip

1. You hold the grip at a steady force, about half your strongest squeeze.
2. The working fibers begin to fatigue, and each one produces less force.
3. To keep the total force steady, the nervous system recruits more motor units and fires them faster.
4. More active fibers make the surface EMG grow larger, even though the force is flat.
5. When no more motor units can be added, the force falls no matter how hard you try.

Oxygen use from rest, through exercise, into recovery

1. At rest, oxygen use is steady at the resting level.
2. When exercise starts, the need for ATP rises at once, but oxygen use rises gradually.
3. Stored ATP, creatine phosphate and anaerobic glycolysis cover the gap. This gap is the oxygen deficit.
4. Oxygen use levels off, and oxidative phosphorylation supplies nearly all the ATP.
5. When exercise stops, oxygen use falls slowly and stays above resting for a while. This is excess post-exercise oxygen consumption (EPOC).
6. The extra oxygen rebuilds ATP and creatine phosphate, reloads myoglobin and hemoglobin, turns lactate back into fuel or glucose, and pays for the higher body temperature and the harder-working heart and breathing muscles.

ATP sources during activity

In your textbook: Silverthorn Chapter 13, the part on how muscle fibers get the ATP for contraction.

A muscle fiber spends ATP on three jobs: crossbridge cycling, pumping Ca2+ back into the sarcoplasmic reticulum with SERCA, and restoring ion gradients with the Na+/K+ pump. It stores only a few seconds' worth of ATP, so it relies on three ways of making more.

Creatine phosphate (phosphocreatine)

A small molecule in muscle that carries a high-energy phosphate. Creatine kinase transfers the phosphate to ADP, making ATP in one step. Fastest source, no oxygen needed, but lasts only seconds. Refilled at rest using ATP from the mitochondria.

Anaerobic glycolysis

Glucose split into two pyruvate in the cytosol, making 2 ATP per glucose. Pyruvate is turned into lactate, which regenerates NAD+. Fast, no oxygen needed, lasts about a minute of hard work.

Oxidative phosphorylation (aerobic metabolism)

ATP made in the mitochondria using oxygen, about 30 to 32 ATP in all from each glucose. Can also use fatty acids and amino acids. Slowest to start and lowest top rate, but lasts as long as oxygen and fuel last. Supplies nearly all the ATP for resting and moderately active muscle.

Source	How fast it supplies ATP	How long it can keep going	Needs oxygen?
Stored ATP and creatine phosphate	Fastest	Seconds, roughly the first 10 to 15	No
Anaerobic glycolysis	Fast	About a minute of hard work	No
Oxidative phosphorylation	Slowest	Hours, while oxygen and fuel last	Yes

- Rate (the tap) and capacity (the tank) are separate properties, but they run in the same order: the fastest source is the first to run out.
- A 100 meter dash runs mostly on stored ATP and creatine phosphate, with glycolysis coming up quickly. An hour-long run depends almost entirely on oxidative phosphorylation.
- Lactate is not wasted. It leaves the fiber and can be used as fuel by other tissues or turned back into glucose by the liver.

Three kinds of muscle fiber

In your textbook: Silverthorn Chapter 13, the part on the classification of skeletal muscle fibers by contraction speed and resistance to fatigue.

Skeletal muscle fibers are classified by how fast they contract and how well they resist fatigue.

Property	Slow oxidative (type I)	Fast oxidative-glycolytic (type IIA)	Fast glycolytic (type IIX)
Speed of contraction	Slow	Fast	Fast
Fiber diameter	Small	Medium	Large
Color	Dark red	Red	Pale (white)
Myoglobin	High	High	Low
Mitochondria	Many	Many, fewer than slow oxidative	Few
Capillaries	Many	Many	Few
Main ATP source	Oxidative phosphorylation	Oxidative and glycolytic	Anaerobic glycolysis
Fatigue resistance	Resists fatigue	Intermediate	Tires quickly
Typical task	Posture, held for hours	Walking, moderate sustained work	Jumping, sprinting, lifting a heavy load

Myoglobin

A red protein in muscle that binds oxygen and helps move it into the fiber. It makes oxidative fibers red.

Fast-twitch fibers

Fast oxidative-glycolytic and fast glycolytic fibers. Their myosin splits ATP faster, and they pump Ca2+ back into the sarcoplasmic reticulum faster, so their twitches rise and fall sooner.

Slow-twitch fibers

Slow oxidative fibers. Slower myosin and slower Ca2+ uptake, so a longer twitch.

- A small diameter keeps the distance oxygen must travel from the capillaries short, which helps a slow oxidative fiber rely on oxidative phosphorylation.
- Every muscle is a mix of fiber types. All the fibers in one motor unit are the same type.
- The proportions differ between muscles and between people, set partly by genetics and partly by use.
- Small motor units of slow oxidative fibers are recruited first for light work. The large motor units of fast glycolytic fibers are added only when a task needs a lot of force.

Fatigue

In your textbook: Silverthorn Chapter 13, the part on muscle fatigue.

Muscle fatigue is a drop in the force a muscle can produce or keep up despite continued effort. It is reversed by rest.

Central fatigue

Fatigue that starts in the central nervous system, the brain and spinal cord. It includes feeling tired and wanting to stop, and it often sets in before the muscles themselves fail, which may help protect the body.

Peripheral fatigue

Fatigue that starts anywhere from the neuromuscular junction to the crossbridges: at the junction, in the action potential along the T tubules, in Ca2+ release from the sarcoplasmic reticulum, or at the crossbridges.

Contributors to peripheral fatigue

- K+ leaves the fiber with each action potential and builds up in the T tubules, changing the membrane potential and reducing Ca2+ release.
- Inorganic phosphate (Pi) builds up as ATP and creatine phosphate are broken down. It interferes with the power stroke and with Ca2+ release, and it is thought to be one of the main causes of fatigue in short bursts of maximal effort.
- Less Ca2+ is released from the sarcoplasmic reticulum, so fewer crossbridges form.
- In long, moderate exercise, fatigue goes along with using up the fiber's glycogen.
- Which factor matters most depends on the kind and intensity of the exercise.
- ATP does not run out completely: the fiber slows down before its ATP is gone.
- Lactic acid (lactate and H+) is no longer thought to be the main cause of muscle fatigue. The fall in pH may still play some part, and research on it continues.

Grip force and EMG (Competency 15, done in lab)

- A grip dynamometer records grip force. Surface EMG electrodes on the forearm record the combined electrical activity of the muscle fibers underneath.

- During a steady submaximal hold, force stays flat while EMG amplitude grows, because more motor units are recruited and fired faster to replace fatiguing ones.
- When recruitment can rise no further, force falls.
- To tell central from peripheral fatigue, the motor nerve can be stimulated electrically once a person can no longer squeeze hard. If force comes back, the failure was central. If force stays low, the fatigue is in the muscle.

Oxygen use after exercise

In your textbook: Silverthorn Chapter 13, the part on muscle fatigue and oxygen, and Chapter 25, the part on oxygen consumption during and after exercise.

Oxygen deficit

The gap at the start of exercise between the oxygen the work needs and the oxygen actually used. Stored ATP, creatine phosphate and anaerobic glycolysis cover it while breathing, heart rate and blood flow rise.

Excess post-exercise oxygen consumption (EPOC)

Oxygen use that stays above resting after exercise ends. It used to be called oxygen debt. It is usually larger than the oxygen deficit, because it pays for more than replacing the deficit.

What the extra oxygen after exercise pays for

- Rebuilding ATP and creatine phosphate through oxidative phosphorylation.
- Reloading myoglobin in the muscles and hemoglobin in the blood with oxygen.
- Turning lactate back into pyruvate to use as fuel, or back into glucose in the liver.
- A higher body temperature, which speeds up metabolism.
- The heart and breathing muscles, which are still working harder than at rest.
- After a sprint, recovery leans more on rebuilding creatine phosphate and clearing lactate.

Training and disuse

In your textbook: Silverthorn Chapter 13, the parts on muscle fiber types and on how exercise changes muscle.

Muscle fibers change with how they are used. This is called plasticity.

Change	Fiber size	Myofibrils	Mitochondria	Capillaries
Months of endurance training	Little change	Little change	Many more	More
Months of resistance training	Much larger (hypertrophy)	Many more	Do not keep pace, spread over a larger area	Do not keep pace, spread over a larger area
Six weeks in a cast (disuse)	Smaller (atrophy)	Fewer	Fewer	Fewer

Hypertrophy

An increase in fiber size, mainly from building more myofibrils with more actin and myosin. The number of fibers changes little.

Disuse atrophy

Shrinking of fibers that are hardly used: more protein is broken down than built. The muscle becomes weaker and tires sooner. It can be rebuilt with use as long as the nerve supply is intact.

- Endurance training also raises myoglobin, and some fast glycolytic fibers become more oxidative, behaving more like fast oxidative-glycolytic fibers.
- Strength depends mostly on how many myofibrils lie side by side. Endurance depends on mitochondria and blood supply. A resistance-trained fiber is the stronger one, and an endurance-trained fiber keeps going longer.
- What this means for you: people on bed rest or in a cast lose muscle quickly, which is one reason early movement and physical therapy are part of recovery. Astronauts in low gravity face the same problem.

BIO 005 Human Physiology · Week 7 notes · Dr. Sharilyn Rennie

How movement gets commanded

The written notes for the how movement gets commanded walkthrough. Week 7 competencies 17 to 20.

To print: these notes as a two-column PDF. Already printed the Week 7 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the walkthrough.

Step by step: from command to contraction

In your textbook: Silverthorn Chapter 14, the parts on the integration of muscle movement in the CNS and on voluntary movements.

Reaching for a glass: the stages of a voluntary movement

1. Decide: the association areas of the cortex decide to reach for the glass.
2. Plan: the association areas work out the plan together with the basal ganglia and the cerebellum. The basal ganglia and the cerebellum send their output back to the cortex by way of the thalamus.
3. Start: the primary motor cortex, the precentral gyrus of the frontal lobe, starts the movement.
4. Carry out: the command travels down the corticospinal tract, and the brainstem pathways adjust posture, all ending on lower motor neurons in the spinal cord. The lower motor neurons fire and the arm and hand muscles contract.
5. Adjust: muscle spindles, tendon organs and joint receptors send feedback up to the cerebellum and the cortex. The cerebellum compares the movement with the intended one and sends corrections to the motor cortex (through the thalamus) and to the brainstem while the movement is still going on.

The corticospinal pathway, neuron by neuron

1. Upper motor neuron: its cell body is a pyramidal cell in the primary motor cortex, the precentral gyrus, just in front of the central sulcus.
2. Its axon runs down through the internal capsule, a narrow band of white matter between the thalamus and the basal ganglia.
3. It continues down through the brainstem: the midbrain, the pons and the medulla.
4. At the bottom of the medulla, in the pyramids, about 85 to 90 percent of the axons cross the midline: the pyramidal decussation.
5. The crossed axons descend on the opposite side of the cord in the lateral corticospinal tract. The few that do not cross descend in the anterior corticospinal tract.
6. Synapse: in the ventral horn, on a lower motor neuron, directly or through an interneuron.
7. Lower motor neuron: its cell body is in the ventral horn. Its axon leaves through the ventral root, runs in the spinal nerve, and releases acetylcholine at the neuromuscular junction. The muscle contracts.

The cerebellum correcting a reach

1. As the reach starts, the left motor cortex sends a copy of its command down to the pons, which passes it across to the right half of the cerebellum.
2. The moving right arm sends feedback from its muscle spindles, tendon organs and joints up the cord to the right half of the cerebellum. The inner ear adds information about balance.
3. The cerebellum compares the intended movement with the actual one.
4. When they do not match, it sends a correction to the left motor cortex, by way of the thalamus, and to the brainstem.
5. The command is adjusted while the hand is still moving, so it slows and stops on the glass.

The basal ganglia loop

1. The cortex sends its possible movements to the basal ganglia.
2. The basal ganglia, supplied with dopamine by the substantia nigra in the midbrain, release the brake on the chosen movement and keep holding back the others. They work mostly by inhibition.
3. Their output goes to the thalamus, and the thalamus passes it back up to the motor cortex.
4. The motor cortex sends the chosen movement down the corticospinal tract.

Levels of motor control

In your textbook: Silverthorn Chapter 14, the part on the integration of muscle movement in the CNS.

Level	What it contributes to a voluntary movement
Spinal cord (lowest)	Runs spinal reflexes on its own, and holds the lower motor neurons that every command must pass through. Holds networks for rhythmic movements such as stepping (central pattern generators).
Brainstem (middle)	Keeps posture and balance through postural reflexes, with information from the inner ear, the eyes and the body. Its own pathways run down to the lower motor neurons.
Cerebral cortex (top)	Decides on, plans and starts voluntary movements. The primary motor cortex sends the command down the corticospinal tract.
Cerebellum (beside)	Compares the intended movement with feedback and corrects it. Coordinates timing and force, and learns skills. Does not send commands down to the muscles.
Basal ganglia (beside)	Select and start the wanted movement and hold back unwanted ones, in a loop back to the cortex through the thalamus. No pathway of their own to the spinal cord.

- A reflex can run in the spinal cord alone. A voluntary movement uses all three levels, with the cerebellum and the basal ganglia shaping it.
- Only the motor cortex and the brainstem send commands down the cord, and both end on the lower motor neurons.
- Cutting the cord in the neck, above the segments for the arm, takes away voluntary movement and the brainstem's control of posture below the cut. Once the period of spinal shock right after the injury passes, spinal reflexes below the cut, such as withdrawal from a pinprick, still work and are often brisker than normal.

Spinal shock

The loss of reflexes below a spinal cord injury for days to weeks right after it happens.

Central pattern generator

A network of neurons in the spinal cord that produces a rhythmic movement, such as the stepping of walking. The brain starts, stops and adjusts it.

The corticospinal pathway

In your textbook: Silverthorn Chapter 14, the part on voluntary movements, and Chapter 10, the part on the motor cortex.

- The primary motor cortex is a map of the body, like the touch map just behind it, with large areas for the hands and the face.
- Two neurons carry the command: the upper motor neuron, from the cortex to the cord, and the lower motor neuron, from the ventral horn to the muscle.
- The axons are packed tightly in the internal capsule, so a small stroke there can weaken the whole opposite side of the body.
- Because of the pyramids, the corticospinal tract is also called the pyramidal tract.
- The lower motor neuron is the final common path: every signal that makes a skeletal muscle contract, from a reflex, the brainstem or the cortex, passes through it. It adds up its excitatory and inhibitory inputs and fires only if the sum reaches threshold. If it is lost, the muscle cannot contract for any reason.

Where the damage is	Side of the body that is weak	Why
Left motor cortex (stroke)	Right	Above the crossing: these fibers would have crossed to the right
Left medulla, just above the place where the pyramids cross	Right	Still above the crossing
Left half of the spinal cord	Left	Below the crossing: these fibers came from the right cortex and already crossed

- Damage above the crossing weakens the opposite side of the body; damage below it weakens the same side.
- Weakness of one side of the body is hemiparesis; complete paralysis of one side is hemiplegia.
- A cortex stroke and a low medulla lesion both weaken the opposite side. The face helps tell them apart: a cortex stroke often weakens the face too, while low medulla damage usually spares it, because the fibers to the face muscles leave the pathway higher in the brainstem.
- Half the cord cut gives the Brown-Séquard pattern from Week 5: weakness and loss of touch and position sense on the same side, loss of pain and temperature on the other side.

Cerebellum and basal ganglia

In your textbook: Silverthorn Chapter 14, the part on voluntary movements, and Chapter 10, the parts on the cerebellum and the basal ganglia.

	Cerebellum	Basal ganglia
Where	Back of the brain, below the cerebrum, behind the brainstem	Clusters of gray matter deep in each half of the brain, beside the thalamus; the substantia nigra is in the midbrain
Job	Compares the intended movement with feedback and corrects it; timing, coordination, balance, learning skills; each half works with the same side of the body	Select and start the wanted movement and hold back unwanted ones
Input	A copy of the command from the cortex, through the pons; feedback from muscles, tendons and joints; balance from the inner ear	Plans from the cortex; dopamine from the substantia nigra
Output	Back to the motor cortex through the thalamus, and to the brainstem	Back to the motor cortex through the thalamus
Tremor when damaged	Intention tremor: during a movement toward a target, worst near the end; gone at rest	Resting tremor (Parkinson disease): at rest, eases when the hand moves
Other signs	Overshoot (dysmetria); clumsy, unsteady movements and walking (ataxia)	Slow to start; slow, small movements (bradykinesia); stiffness (rigidity); shuffling steps, little facial expression

Intention tremor

A tremor that appears during a movement toward a target and grows near the end. Points to the cerebellum.

Dysmetria

Misjudging the distance of a movement, so the hand overshoots or falls short.

Ataxia

Clumsy, poorly coordinated movement, including an unsteady walk.

Resting tremor

A tremor at rest that eases with movement. Points to the basal ganglia, as in Parkinson disease.

Bradykinesia

Slowness of movement.

Rigidity

Stiffness felt when someone else moves the limb.

- Basal ganglia that let too little through: movements hard to start, slow and small (Parkinson disease). Too much: unwanted movements break out (Huntington disease, with jerky, dance-like movements).
- Parkinson disease: the dopamine neurons of the substantia nigra die. Treatment is L-dopa, a precursor that crosses the blood-brain barrier and is made into dopamine in the brain; dopamine itself cannot cross.
- Alcohol affects the cerebellum early, which is why roadside tests ask for walking heel to toe or touching the nose. Years of heavy drinking can permanently damage the cerebellum and leave an unsteady walk.
- When a tremor appears tells you the structure: during movement for the cerebellum, at rest for the basal ganglia.

Upper and lower motor neuron signs

In your textbook: Silverthorn Chapter 14, the part on voluntary movements.

Finding	Upper motor neuron damage	Lower motor neuron damage
Strength	Weak	Weak
Tone	High: stiff, resists fast stretch (spasticity)	Low: floppy (flaccid)
Tendon reflexes	Brisk, often with clonus	Weak or absent
Muscle bulk	Little wasting at first; slow loss from disuse	Marked wasting (atrophy)
Twitches at rest	None	Fasciculations
Sole stroke	Big toe goes up (Babinski sign)	No Babinski sign
Examples	Stroke, spinal cord injury above that level	Polio, a crushed peripheral nerve

- Weakness alone does not locate the damage. Reflexes are a good first check, because they go in opposite directions.
- Upper motor neuron damage: the reflex arc is intact but the brain's pathways that hold it back are gone, so reflexes are brisk. Right after a stroke or cord injury the limb can be floppy with weak reflexes for a while; stiffness and brisk reflexes develop over days to weeks.
- Lower motor neuron damage: the last step of the reflex arc is broken, so the reflex weakens or disappears, and the muscle loses its nerve supply and wastes.
- Polio destroys ventral horn motor neurons: weak, floppy, wasted muscles with sensation spared. A single injured nerve gives the same signs only in the muscles it supplies.
- ALS (amyotrophic lateral sclerosis): upper and lower motor neurons die, sensation is spared. Wasting and fasciculations appear together with brisk reflexes and sometimes a Babinski sign, in more than one part of the body. Speech, swallowing and breathing muscles weaken; breathing failure is the most common cause of death.

Putting it together: a patient with right-sided weakness

1. Brisk reflexes and a Babinski sign on the right: upper motor neuron damage.
2. Weakness on the right: if the damage is above the crossing, it is on the left side.
3. The right side of the face droops too: the damage is in the brain, not the cord, because the fibers to the face never go down into the cord.
4. Answer: an upper motor neuron lesion in the left side of the brain, such as a stroke in the left motor cortex or internal capsule.